

Topological superconductors and Majorana fermions based on X -wave magnets with $X = p, d, f, g, i$

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We study superconductors coupled with X -wave magnets with $X = p, d, f, g, i$. They are essentially different between altermagnets with $X = d, g, i$ and odd-parity magnets $X = p, f$ due to the compatibility of the spin-singlet formation of the s -wave superconductor. The altermagnets and odd-parity magnets show rich superconducting phase diagrams depending on the strength of the magnetic order, the chemical potential and the ratio of the on-site and nearest-neighbor interactions. Especially, we find that the altermagnetic order stabilizes the chiral $(p_x + ip_y)$ -wave superconductor, which is a class D topological superconductor hosting Majorana chiral edge states. It is characterized by the Chern number. Especially, there emerge two chiral edge modes in the i -wave altermagnet, which are characterized by the Chern number 2. On the other hand, the p (f)-wave odd-parity magnetic order stabilizes the p (f)-wave topological superconductor, which is a class DIII topological superconductor hosting Majorana flat bands. It is characterized by the winding number. Majorana flat bands are robust in the presence of the mixing with the s -wave superconductivity, which is inevitable for odd-parity magnets although the magnitude of the mixing is tiny.

I. INTRODUCTION

Superconductivity is one of the most fascinating phenomena in condensed-matter physics. A conventional superconductor has the s -wave superconducting gap, which is isotropic and has a full gap. On the other hand, d -wave superconductivity is well studied in the context of the high-temperature superconductivity. In addition, $p_x + ip_y$ superconductors are known to be topological superconductors[1], which have Majorana chiral-edge states.

Altermagnets[2–9] and odd-parity magnets[10–20] are new-types of magnets with zero-net magnetization, which are characterized by the spin-split electronic band structure. They are protected by a combinatorial symmetry of the spin flip and the momentum rotation. Altermagnets constitute of d -wave, g -wave and i -wave magnets, while odd-parity magnets constitute of p -wave and f -wave magnets. They are summarized as X -wave magnets[21–23] with $X = p, d, f, g, i$. The interplay between superconductivity and X -wave magnets is interesting because there is no net magnetization in the X -wave magnet. It is shown that the chiral $p_x + ip_y$ -wave topological superconductor is stabilized when it is coupled with the d -wave altermagnet[24]. On the other hand, the Majorana flat bands emerge when it is coupled with the p -wave magnet[25], where there is a mixing of the p_x -wave and s -wave superconducting gaps. Although superconductors coupled with d -wave altermagnets[26–36] and p -wave magnets[25, 37–42] are studied, those coupled with g -wave altermagnets, i -wave altermagnets and f -wave altermagnets have been scarcely studied.

In this paper, we study superconductors coupled with X -wave magnets. We find that the phase diagrams are essentially different between altermagnets and odd-parity magnets. The difference arises from the fact that spin-singlet superconducting pairing is compatible with the altermagnetic order, while it is not in odd-parity magnets. Especially, the p -wave chiral topological superconductor is realized in the case of altermagnets. Furthermore, chiral $(p_x + ip_y)$ supercon-

ductivities emerge universally in the case of the d -wave, g -wave and i -wave altermagnets. They are class D topological superconductors[43] due to the time-reversal breaking of altermagnets. There emerge chiral edge modes, which are characterized by the Chern number. Especially, there emerge two chiral edge modes in the i -wave altermagnet, which are characterized by the Chern number 2. On the other hand, there emerge Majorana flat bands in the p -wave (f -wave) superconductor coupled with the p -wave (f -wave) magnet, which is characterized by the modulo 2 of the winding number. They are class DIII topological superconductors[43] because odd-parity magnets preserve time-reversal symmetry in the electronic band structure.

II. RESULTS

A. Superconducting gap

We study the extended Hubbard model with on-site and nearest-neighbor attractive interactions,

$$H_{\text{Hubbard}} = -t \sum_{\langle j, j' \rangle, s} c_{js}^\dagger c_{j's} - U \sum_j c_{j\uparrow}^\dagger c_{j\uparrow} c_{j\downarrow}^\dagger c_{j\downarrow} - V \sum_{\langle j, j' \rangle} n_j n_{j'} \quad (1)$$

with the density

$$n_j = \sum_s c_{js}^\dagger c_{js}, \quad (2)$$

where t denotes the hopping amplitude, U is the strength of the on-site attractive interaction strength, and V represents the attractive interaction strength between nearest-neighbor sites. $\langle j, j' \rangle$ represents the nearest-neighbor sites j and j' , and $c_{js}^\dagger$ (c_{js}) represents the creation (annihilation) operator of electrons with spin $s = \uparrow, \downarrow$. We make an analysis based on the square lattice for d -wave altermagnets, g -wave altermagnets

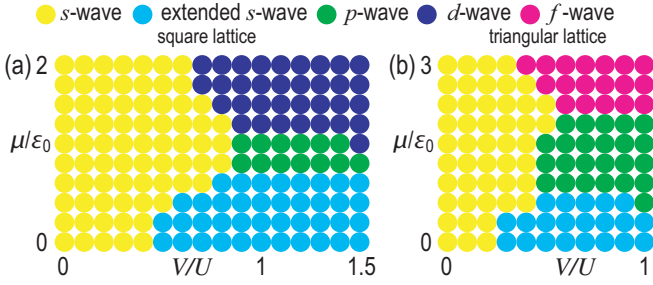


FIG. 1. Phase diagram with $J = 0$. (a) Square lattice and (b) triangular lattice. The horizontal axis is V/U and the vertical axis is μ/ε_0 . Yellow disks represent the s -wave superconductivity, cyan disks represent the extended s -wave superconductivity, green disks represent the in-plane chiral p -wave superconductivity in (a) and out-of-plane p_x -wave superconductivity in (b), blue disks represent the d -wave superconductivity, and magenta disks represent the f -wave superconductivity. We have set $t = \varepsilon_0/2$ and $U = \varepsilon_0$, where ε_0 is the unit of energy.

and p -wave magnets, while the triangular lattice for i -wave altermagnets and f -wave magnets.

Due to the Pauli exclusion principle, spin-singlet superconducting gaps must satisfy

$$\Delta(-\mathbf{k}) = \Delta(\mathbf{k}), \quad (3)$$

while spin-triplet superconducting gaps must satisfy

$$\Delta(-\mathbf{k}) = -\Delta(\mathbf{k}). \quad (4)$$

Spin-singlet superconducting gaps have s -wave, d -wave, g -wave and i -wave symmetries, which have even parities. On the other hand, spin-triplet superconducting gaps have p -wave and f -wave symmetries, which have odd parities. We note that g -wave and i -wave superconducting gaps cannot be stabilized in the Hamiltonian (1) because they require longer-range interactions. Furthermore, f -wave superconducting gap is realized only in the triangular lattice.

We study the superconductivity based on the Bogoliubov-de Gennes (BdG) Hamiltonian,

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \hat{\xi}(\mathbf{k}) & \Delta(\mathbf{k}) \\ \Delta^\dagger(\mathbf{k}) & -\hat{\xi}^T(-\mathbf{k}) \end{pmatrix}, \quad (5)$$

where $\hat{\xi}(\mathbf{k})$ is the kinetic energy. We solve the linearized gap equation and calculate which superconductor has the highest critical temperature, and determine the phase diagram. Details are shown in Methods IV A.

B. Phase diagram without magnetic order

We start with a review of the phase diagram of superconductors without the X -wave magnet. When the on-site interaction is dominant, the s -wave superconductor is realized. On the other hand, the phase diagram shows a rich variety when the nearest-neighbor interaction is dominant. It depends on whether the lattice is square or triangular.

We first study the square lattice, whose phase diagram is shown in Fig. 1(a). The extended s -wave superconductor induced by V is realized in the vicinity of the band bottom, where the Fermi surface is almost a circle. See Eq.(35) for the definition of the gap function of the extended s -wave superconductivity. The chiral p -wave superconductor is realized when the chemical potential is near the 1/4 filling. The d -wave superconductor is realized at the half filling.

We next study the triangular lattice, whose phase diagram is shown in Fig. 1(b). The extended s -wave superconductor is realized for $0.3 \lesssim V/U \lesssim 0.6$, while the chiral p -wave superconductor is realized for $V/U \gtrsim 0.6$ in the vicinity of the band bottom. The chiral p -wave superconductor is realized when the chemical potential is near the 1/4 filling. The f -wave superconductor is realized when V is dominant. The phase boundary V/U between the s -wave superconductor and the extended s -wave superconductor in the triangular lattice is smaller than that in the square lattice. It is because the number of the adjacent site is six in the triangular lattice and it is four in the square lattice, where the effective next-nearest neighbor interactions in the triangular lattice are stronger than those in the square lattice.

C. X -wave magnets

The electromagnetic property of the X -wave magnet is characterized by the peculiar band splitting depending on the spin. The simplest model is the two-band Hamiltonian[9, 21–23, 49] given by[24, 25, 35, 36, 42]

$$H_X = H_{\text{Hubbard}}(\mathbf{k}) + J f_X(\mathbf{k}) \sigma_z, \quad (6)$$

where the first term $H_{\text{Hubbard}}(\mathbf{k})$ represents the Hamiltonian (1), while the second term represents the band splitting described by the function $f_X(\mathbf{k})$ with the coupling constant J and the Pauli matrix σ_z . Explicit forms of $f_X(\mathbf{k})$ are given in Methods IV D. We have $f_X(-\mathbf{k}) = f_X(\mathbf{k})$ for altermagnets, while $f_X(-\mathbf{k}) = -f_X(\mathbf{k})$ for odd-parity magnets. As a result, the linearized gap equation for altermagnets reads

$$-\frac{V_{\alpha\alpha} k_B T}{N_\Gamma} \sum_{\mathbf{k}} \frac{\Gamma_\alpha^\dagger \Gamma_\alpha}{4H_{\text{kin}}(\mathbf{k}) k_B T} \sum_{s=\pm 1} \tanh \frac{\varepsilon_s(\mathbf{k})}{2k_B T} = 1 \quad (7)$$

for the singlet superconducting gap and the out-of-plane triplet superconducting gap, while it reads

$$-\frac{V_{\alpha\alpha} k_B T}{N_\Gamma} \sum_{\mathbf{k}} \frac{\Gamma_\alpha^\dagger \Gamma_\alpha}{4k_B T} \sum_{s=\pm 1} \frac{\tanh \frac{\varepsilon_s(\mathbf{k})}{2k_B T}}{\varepsilon_s(\mathbf{k})} = 1 \quad (8)$$

for the in-plane triplet superconducting gap. For odd-parity magnets, Eq.(8) is the gap equation for the singlet superconducting gap and the out-of-plane triplet superconducting gap, while Eq.(7) is the gap equation for the in-plane triplet superconducting gap. Namely, the gap equations are reversed for the altermagnets and the odd-parity magnets. See details in Methods IV A.

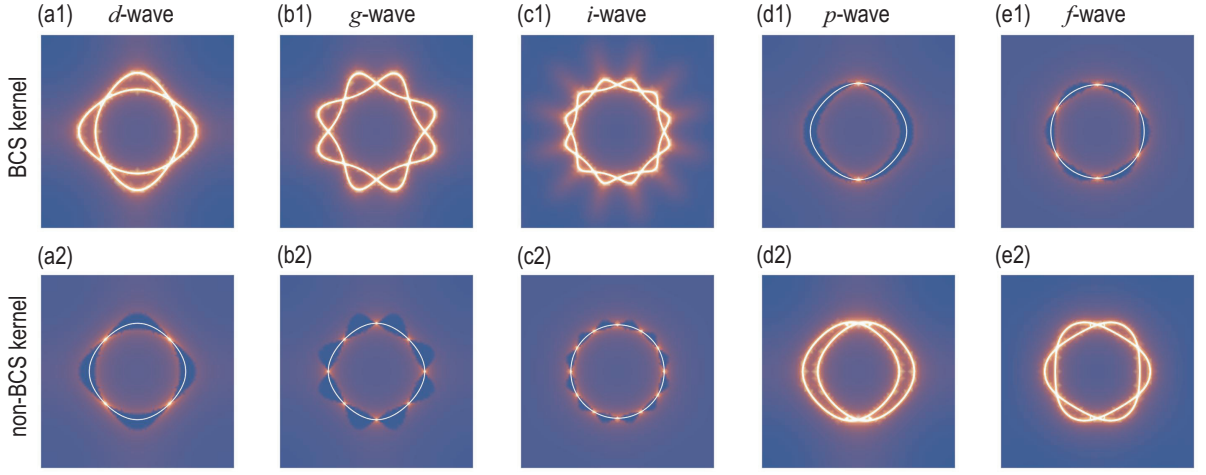


FIG. 2. (a1)~(e1) Momentum distribution of the BCS kernel $K_{\text{BCS}}(\mathbf{k})$. (a2)~(e2) Momentum distribution of the non-BCS kernel $K_{\text{BCS}}^{\text{no}}(\mathbf{k})$. The horizontal axis and the vertical axis is k_x and k_y , respectively. We have set $\mu/\varepsilon_0 = 1$.

D. Altermagnets

Altermagnets break time-reversal symmetry. It leads to the class D topological superconductors with the $p_x + ip_y$ chiral superconducting gap, which is a full gap. We show that there emerge chiral edge states, which are characterized by the Chern number.

The critical temperature of the solution of Eq.(7) parabolically decreases as the increase of J as

$$T_{\text{cr}} = T_{\text{cr}}^{(0)} - 28\zeta'(-2) \frac{J^2 \langle f(\mathbf{k})^2 \rangle_{\text{FS}}}{T_{\text{cr}}^{(0)}}, \quad (9)$$

where $\langle f(\mathbf{k})^2 \rangle_{\text{FS}}$ is the integration over the Fermi surface, and

$$\zeta'(x) \equiv \frac{d\zeta(x)}{dx} \quad (10)$$

is the derivative of the zeta function. See the derivation in Methods IV C. On the other hand, the critical temperature of the solution of Eq.(8) is almost irrelevant to J .

This peculiar behavior of the critical temperature (9) is intuitively understood as follows. The singlet superconducting gap and the out-of-plane triplet superconducting gap are formed by the Cooper pair of electron with opposite spins. The electrons with $\mathbf{k}$ and $-\mathbf{k}$ have the same spin in the case of the altermagnet, while the electrons with $\mathbf{k}$ and $-\mathbf{k}$ have opposite spins in the case of odd-parity magnets. It contradicts with the singlet superconducting gap and the out-of-plane triplet superconducting gap. Hence, their critical temperature is lowered. On the other hand, the in-plane triplet superconducting gap is formed by the same spins. Hence, it does not contradict with the altermagnetic spin splitting and their critical temperature is scarcely affected, while it contradicts with the odd-parity magnetic spin splitting and their critical temperature is lowered.

It is also mathematically understood as follows. Eq.(8) contains the BCS kernel

$$K_{\text{BCS}}(\mathbf{k}) = \sum_{s=\pm 1} \frac{\tanh \frac{\varepsilon_s(\mathbf{k})}{2k_{\text{B}}T}}{4k_{\text{B}}T\varepsilon_s(\mathbf{k})}. \quad (11)$$

It is identical to the essential term contained in the well-known BCS equation in the absence of the magnetic order, where the zero of the denominator and the zero of the eliminator perfectly coincide. We show $K_{\text{BCS}}(\mathbf{k})$ in Fig.2(a), where all of the Fermi sea contribute to the superconductivity. Hence, the critical temperature in the presence of magnetic order is almost identical to that in the absence of the magnetic order. On the other hand, Eq.(7) contains the term

$$K_{\text{BCS}}^{\text{no}}(\mathbf{k}) = \sum_{s=\pm 1} \frac{\tanh \frac{\varepsilon_s(\mathbf{k})}{2k_{\text{B}}T}}{4k_{\text{B}}TH_{\text{kine}}(\mathbf{k})}. \quad (12)$$

Its integration over $\mathbf{k}$ is much smaller than the integration of Eq.(11) over $\mathbf{k}$. We show $K_{\text{BCS}}^{\text{no}}(\mathbf{k})$ in Fig.2(b), where only the crossing points of the up and down spins have large values. It leads to the lower critical temperature.

Indeed, we will see in the next sections that the s -wave and the extended s -wave and d -wave superconductivities are suppressed for large J for altermagnets, while it is not for odd-parity magnets. On the other hand, the in-plane p -wave superconductivity is not suppressed by the increase of J . Hence, the in-plane p -wave superconductor is realized for sufficiently large J for altermagnets.

1. d -wave altermagnets

The phase diagram in the $(V/U, J)$ plane is shown in Fig.3. First, we study the low chemical potential regime by setting $\mu/\varepsilon_0 = 0.5$. The s -wave conductivity is destroyed for strong J because the spin singlet pairing is not compatible with the

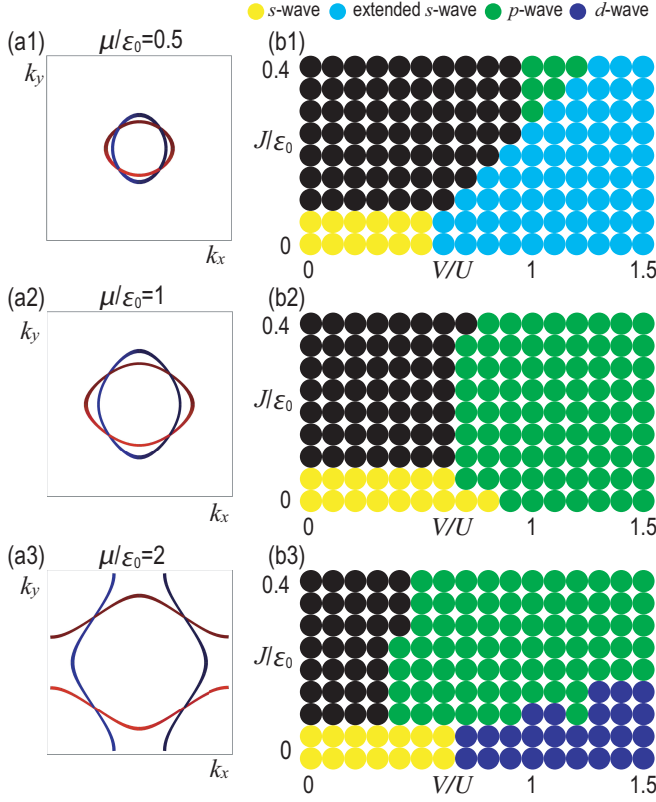


FIG. 3. *d*-wave alternemagnets. (a1), (a2) and (a3) Fermi surface. (b1), (b2) and (b3) Phase diagram. (a1) and (b1) $\mu/\varepsilon_0 = 0.5$, (a2) and (b2) $\mu/\varepsilon_0 = 1$ and (a3) and (b3) $\mu/\varepsilon_0 = 2$. The horizontal axis is V/U , while the vertical axis is J/ε_0 . Black disks represent the absence of superconductivity. We have set $J/\varepsilon_0 = 0.4$ for the Fermi surface. See also the caption of Fig.1 for other colors.

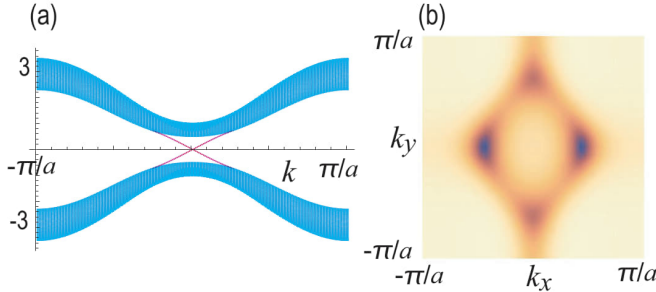


FIG. 4. *d*-wave alternemagnets. (a) Energy spectrum with the $p_x + ip_y$ chiral superconducting gap in nanoribbon geometry for *d*-wave alternemagnets. Red color indicates that the amplitude at the edges is large, while cyan color indicates that it is small. We have set $t = -\varepsilon_0/2$, $\mu = \varepsilon_0/2$, $J = 0.2\varepsilon_0$, $\Delta_d = \varepsilon_0/2$ and $L = 40$. (b) Momentum dependence of the Berry curvature.

d-wave alternemagnetic Fermi surface. The chiral *p*-wave superconductor is realized in the region around $V/U \simeq 1$ and $J/\varepsilon_0 \simeq 0.4$. Next, we study the middle chemical potential regime ($\mu/\varepsilon_0 = 1$). The region of the *p*-wave superconductivity becomes broader around $V/U \lesssim 1$ for strong J because the *s*-wave superconductivity is destabilized coupled with the

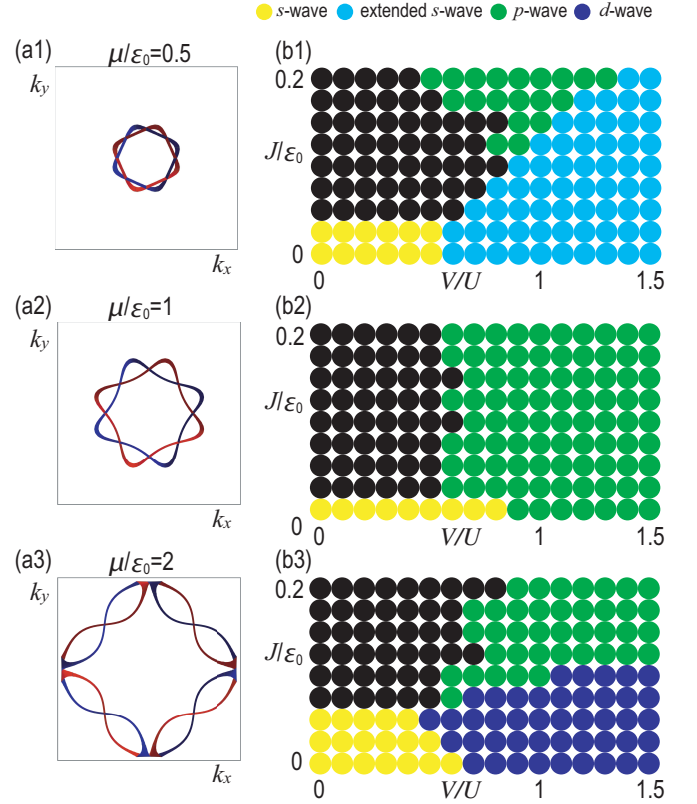


FIG. 5. *g*-wave alternemagnets. (a1)~(a3) Illustration of the Fermi surface. (b1)~(b3) Phase diagram. We have set $J/\varepsilon_0 = 0.1$ for the Fermi surface. See also the caption of Fig.3.

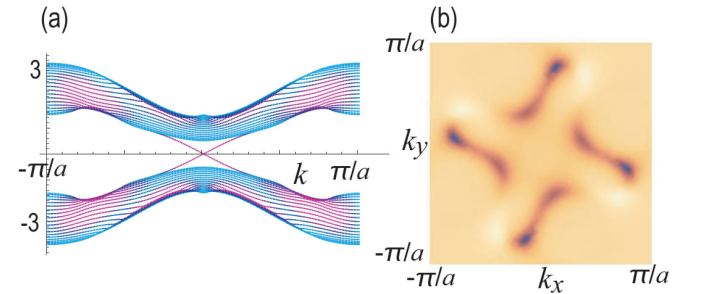


FIG. 6. *g*-wave alternemagnets. (a) Energy spectrum with the $p_x + ip_y$ chiral superconducting gap in nanoribbon geometry for *g*-wave alternemagnets. We have set $t = -\varepsilon_0/2$, $\mu = \varepsilon_0/2$, $J = 0.2\varepsilon_0$, $\Delta_g = \varepsilon_0/2$ and $L = 40$. (b) Momentum dependence of the Berry curvature.

d-wave alternemagnetic order. Finally, we study the phase diagram at the half filling ($\mu/\varepsilon_0 = 2$). The *p*-wave superconductivity is stabilized instead of the *d*-wave superconductivity for strong J . The phase diagram is consistent with the previous result[24].

The band structure in nanoribbon geometry is shown in Fig.4. There are chiral edge states indicating the Majorana flat bands. Correspondingly, the Chern number is evaluated to be 1 based on the numerical integration of Eq.(77) in Methods.

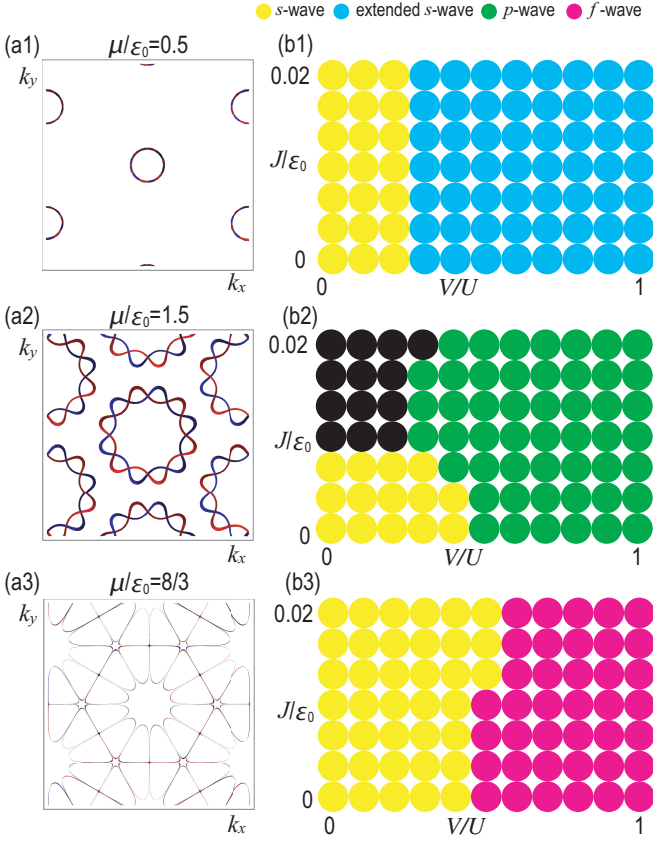


FIG. 7. *i*-wave altermagnets. (a1)~(a3) Illustration of the Fermi surface. (b1)~(b3) Phase diagram. (a1) and (b1) $\mu/\varepsilon_0 = 0.5$, (b1) and (b2) $\mu/\varepsilon_0 = 1.5$, (a3) and (b3) $\mu/\varepsilon_0 = 8/3$. We have set $J/\varepsilon_0 = 0.025$ for the Fermi surface.

2. *g*-wave altermagnets

The phase diagram in the $(V/U, J)$ plane is shown in Fig.5. The feature of the phase diagram is almost identical to that of the *d*-wave altermagnet although the positions of phase boundaries are different.

The band structure in nanoribbon geometry is shown in Fig.6. There are chiral edge states indicating the Majorana flat bands. Correspondingly, the Chern number is evaluated to be 1 based on the numerical integration of Eq.(77) in Methods.

3. *i*-wave altermagnets

The phase diagram in the $(V/U, J)$ plane is shown in Fig.7. The phase diagrams for the low and middle chemical potential regimes are similar to those of the *d*-wave and *g*-wave altermagnets. We study the chemical potential at the van Hove singularity ($\mu/\varepsilon_0 = 8/3$).

The band structure in nanoribbon geometry is shown in Fig.8. There are two chiral edge states indicating the Majorana zero modes. The Chern number is evaluated to be 2 based on the numerical integration of Eq.(77) in Methods.

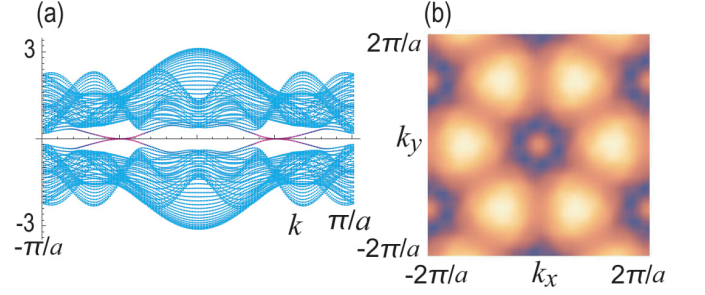


FIG. 8. *i*-wave altermagnets. (a) Energy spectrum with the $p_x + ip_y$ chiral superconducting gap in nanoribbon geometry for *i*-wave altermagnets. We have set $t = -2\varepsilon_0/3$, $\mu = 2\varepsilon_0$, $J = 0.02\varepsilon_0$, $\Delta_i = \varepsilon_0/2$ and $L = 40$. (b) Momentum dependence of the Berry curvature.

E. Odd-parity magnets

Odd-parity magnets preserve time-reversal symmetry. It leads to the class DIII topological superconductors, where $\mathbb{Z}_2$ index is the topological number. In the case of the *X*-wave odd-parity magnet with $X = p$ and *f*, *X*-wave superconductor has point nodes with the 2 nodes and 6 nodes for $X = p$ and *f*, respectively. We show that Majorana flat bands emerge for the momentum k_y between the point nodes. They are characterized by nontrivial winding numbers $W(k_y)$.

1. *p*-wave magnets

The phase diagram in the $(V/U, J)$ plane is shown in Fig.9. The phase boundary is almost independent of J , which is highly contrasted with that of altermagnets. It is because the *s*-wave superconductivity is not destroyed by the presence of the *p*-wave magnetic order.

The band structure of the p_x -wave superconductor in nanoribbon geometry is shown in Fig.10(a). There are flat-band zero-energy edge states indicating the Majorana flat bands. The emergence of the Majorana flat bands well agrees with the winding number as shown in Fig.10(b). The momentum k_y dependent winding number is defined by[25, 44]

$$W(k_y) \equiv \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{q(\mathbf{k})} \frac{\partial q(\mathbf{k})}{\partial k_x} dk_x \quad (13)$$

with

$$q(\mathbf{k}) = 2t(\cos ak_x + \cos ak_y - \mu) + J \sin ak_x + i\Delta_p \sin ak_x. \quad (14)$$

See the details in Methods IVE 2. It is explicitly evaluated as $W(k_y) = 1$ for $|\mu - \cos ak_y| < 1$ and $W(k_y) = 0$ for $|\mu - \cos ak_y| > 1$. The Majorana flat bands appear for the momentum satisfying $W(k_y) = 1$.

Actually, there is a mixing of the *s*-wave and p_x -wave superconductivities[25, 39, 42] due to the off-diagonal term in Eq.(32) in Methods. Even in this case, Majorana flat bands remain as they are although the band crossing momenta change.

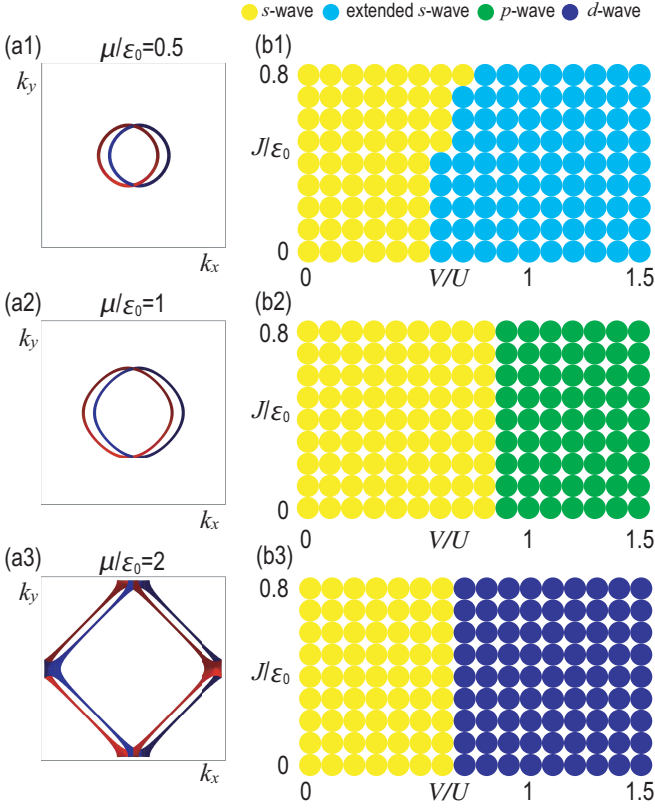


FIG. 9. p -wave magnets. (a1)~(a3) Illustration of the Fermi surface. (b1)~(b3) Phase diagram. See also the caption of Fig. 3. We have set $J/\varepsilon_0 = 0.2$ for the Fermi surface.

This is understood in terms of the winding number (13) of $q_p(\mathbf{k})$ defined by

$$q_p(\mathbf{k}) = 2t(\cos ak_x + \cos ak_y - \mu) + J \sin ak_x + i(\Delta_s + \Delta_p \sin ak_x). \quad (15)$$

The winding number is

$$W(k_y) = 1 \quad (16)$$

for

$$\left(\frac{J\Delta_s}{2t\Delta_p} + \mu - \cos ak_y\right)^2 + \left(\frac{\Delta_s}{\Delta_p}\right)^2 < 1, \quad (17)$$

while

$$W(k_y) = 0 \quad (18)$$

for

$$\left(\frac{J\Delta_s}{2t\Delta_p} + \mu - \cos ak_y\right)^2 + \left(\frac{\Delta_s}{\Delta_p}\right)^2 > 1. \quad (19)$$

See the derivation in Methods IV E 2.

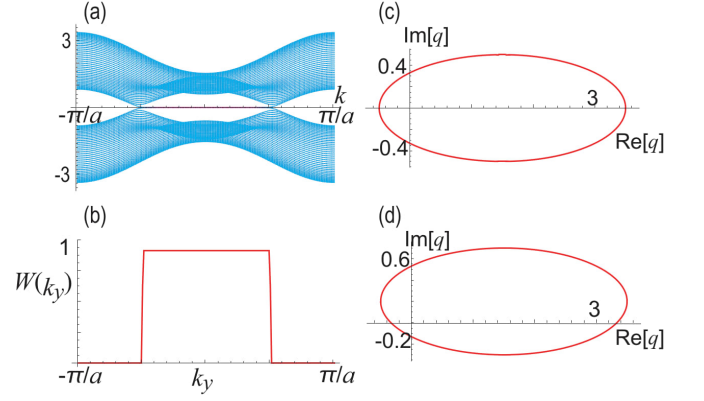


FIG. 10. p -wave magnets. (a) Energy spectrum in nanoribbon geometry for p -wave magnets with $(p_x + s)$ -wave superconductivity. (b) Winding number as a function of k_y , which corresponds to the emergence of Majorana flat bands in (a). (c) Polar plot of $q(\mathbf{k})$ at $k_y = 0$ in the complex plane for the p_x -wave superconductivity. (d) Polar plot of $q(\mathbf{k})$ at $k_y = 0$ in the complex plane for the $(p_x + s)$ -wave superconductivity. We have set $t = -\varepsilon_0/2, \mu = 2\varepsilon_0, J = 0.2\varepsilon_0, \Delta_p = \varepsilon_0/2, \Delta_s = 0.1\varepsilon_0$ and $L = 40$.

2. f -wave magnets

The phase diagram in the $(V/U, J)$ plane is shown in Fig. 11. The phase diagram of the f -wave magnet is similar to that of the p -wave magnet.

The band structure of the f -wave superconductivity with $(p_x + s)$ superconductivity in nanoribbon geometry is shown in Fig. 12(a). There are flat-band zero-energy edge states indicating the Majorana flat bands. The winding number (13) is shown in Fig. 12(b). The winding number takes $W(k_y) = 2, 1$, and -1 . However, the topological number should be $\mathbb{Z}_2$ because the topological class is DIII. Hence, we use $\text{mod}_2 W(k_y)$ as the topological number. Correspondingly, there are no Majorana flat bands around $k_y = \pi/a$, where $W(k_y) = 2$, while there emerge Majorana flat bands for $W(k_y) = \pm 1$. We also show $q_f(\mathbf{k})$,

$$q_f(\mathbf{k}) = 2t \left(\cos ak_x + \cos a \frac{k_x - \sqrt{3}k_y}{2} + \cos a \frac{k_x + \sqrt{3}k_y}{2} \right) - 6t - \mu + J f_f(\mathbf{k}) + i(\Delta_s + \Delta_f f_f(\mathbf{k})) \quad (20)$$

with

$$f_f(\mathbf{k}) \equiv \frac{16}{3\sqrt{3}} \left(\sin ak_x \sin a \frac{k_x + \sqrt{3}k_y}{2} \sin a \frac{k_x - \sqrt{3}k_y}{2} \right) \quad (21)$$

in the complex plane in Fig. 12(c) and (d) for the p -wave superconductivity and $(p_x + s)$ superconductivity, respectively. The topological superconductivity is robust against the mixing of the s -wave superconductivity because the effect of the s -wave superconductivity is the shift of $q(\mathbf{k})$ along the imaginary direction. Indeed, the Majorana flat bands are robust in the presence of the mixing of the s -wave superconductivity as shown in Fig. 12(a).

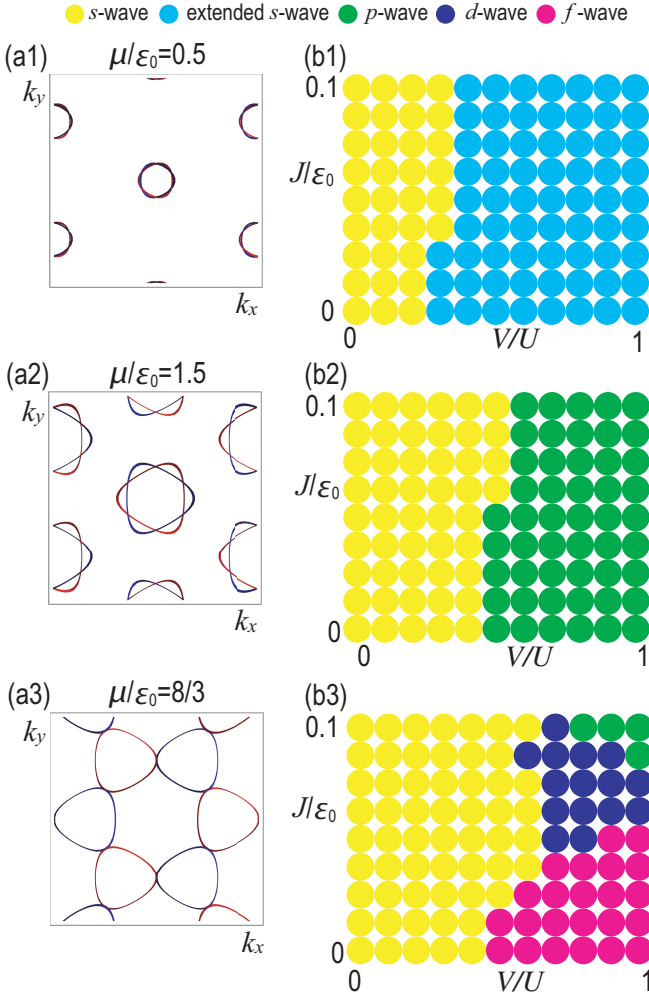


FIG. 11. *f*-wave magnets. Illustration of the Fermi surface. We have set $J/\varepsilon_0 = 0.05$ for the Fermi surface. See also the caption of Fig. 7.

III. DISCUSSION

We have systematically studied the phase diagram of superconductors coupled with the *X*-wave magnet. The phase diagram is essentially different between the altermagnets and the odd-parity magnets. The main reason is that the *s*-wave singlet pairing is not compatible with the altermagnetic order but is compatible with the odd-parity magnetic order. We find class D chiral $p_x + ip_y$ -wave topological superconductors are realized in altermagnets accompanied with chiral edge modes, while class DIII $p + s$ ($f + s$) topological superconductors are realized in *p* (*f*)-wave magnets accompanied with Majorana flat bands between the superconducting gap nodes.

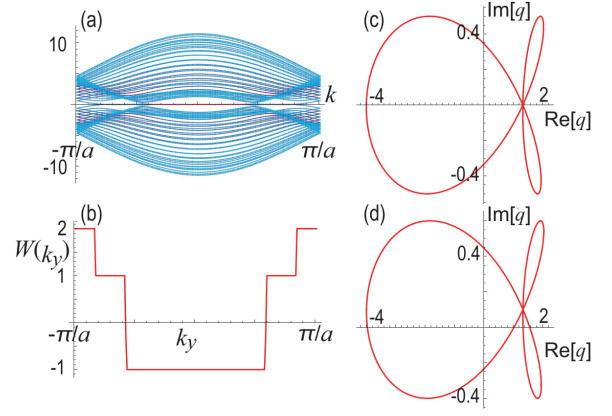


FIG. 12. *f*-wave magnets. Energy spectrum in nanoribbon geometry for *f*-wave magnets with (*f* + *s*)-wave superconductivity. We have set $t = -2\varepsilon_0/3$, $\mu = 8\varepsilon_0/3$, $J = 0.05\varepsilon_0$, $\Delta_f = \varepsilon_0/2$, $\Delta_s = 0.1\varepsilon_0$ and $L = 40$. See also the caption of Fig. 10.

IV. METHODS

A. Linearized gap equations

Superconducting gaps are summarized in terms of the $\mathbf{d}(\mathbf{k})$ vector

$$\Delta(\mathbf{k}) = i(\mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}) \sigma_y. \quad (22)$$

We expand the superconducting gap functions by the orthogonal functions $\Gamma(\mathbf{k})$ as

$$\Delta_m(\mathbf{k}) = \Delta_m \Gamma_m(\mathbf{k}). \quad (23)$$

We also expand the interaction by the function $\Gamma(\mathbf{k})$ as[36]

$$V_{m\ell} = \frac{1}{N^2} \sum_{\mathbf{k}, \mathbf{k}'} \text{Tr} \left[\Gamma_m^\dagger(\mathbf{k}) \hat{V}(\mathbf{k}, \mathbf{k}') \Gamma_\ell(\mathbf{k}') \right], \quad (24)$$

where

$$V(\mathbf{k}, \mathbf{k}') = U + 2V [\cos(k_x - k'_x) + \cos(k_y - k'_y)] \quad (25)$$

for the square lattice[18, 25] and

$$\begin{aligned} V(\mathbf{k}, \mathbf{k}') &= U + 2V [\cos(k_x - k'_x) + 2 \cos \frac{k_x - k'_x}{2} \cos(\frac{\sqrt{3}}{2} (k_y - k'_y))] \\ & \quad (26) \end{aligned}$$

for the triangular lattice.

1. Altermagnets

Linearized gap equations for altermagnets are given by[24, 36]

$$\Lambda_{m'\ell'}(T) = -\frac{V_{m'm'} k_B T}{N_\Gamma} \sum_{\mathbf{k}} M_{m'\ell'}^{\text{even}} \quad (27)$$

with

$$M_{m'\ell'}^{\text{even}} \equiv \begin{pmatrix} \Gamma_0^\dagger \Gamma_0 K_{\text{BCS}}^{\text{no}} & 0 & 0 & 0 \\ 0 & \Gamma_x^\dagger \Gamma_x K_{\text{BCS}} & -i\Gamma_x^\dagger \Gamma_y K_{\text{BCS}}^s & 0 \\ 0 & i\Gamma_y^\dagger \Gamma_x K_{\text{BCS}}^s & \Gamma_y^\dagger \Gamma_y K_{\text{BCS}} & 0 \\ 0 & 0 & 0 & \Gamma_z^\dagger \Gamma_z K_{\text{BCS}}^{\text{no}} \end{pmatrix}, \quad (28)$$

where

$$K_{\text{BCS}}(\mathbf{k}) \equiv \sum_{s=\pm 1} \frac{\tanh \frac{\varepsilon_s(\mathbf{k})}{2k_B T}}{4k_B T \varepsilon_s(\mathbf{k})}, \quad (29)$$

$$K_{\text{BCS}}^s(\mathbf{k}) \equiv \sum_{s=\pm 1} s \frac{\tanh \frac{\varepsilon_s(\mathbf{k})}{2k_B T}}{4k_B T \varepsilon_s(\mathbf{k})}, \quad (30)$$

and

$$K_{\text{BCS}}^{\text{no}} \equiv \sum_{s=\pm 1} \frac{\tanh \frac{\varepsilon_s(\mathbf{k})}{2k_B T}}{4k_B T H_{\text{kin}}(\mathbf{k})}. \quad (31)$$

2. Odd-parity magnets

Linearized gap equations for odd-parity magnets are given by[25]

$$\Lambda_{m'l'}(T) = -\frac{V_{m'm'} k_B T}{N_\Gamma} \sum_{\mathbf{k}} M_{m'l'}^{\text{odd}} \quad (32)$$

with

$$M_{m'l'}^{\text{odd}} \equiv \begin{pmatrix} \Gamma_0^\dagger \Gamma_0 K_{\text{BCS}} & 0 & 0 & \Gamma_0^\dagger \Gamma_z K_{\text{BCS}}^s \\ 0 & \Gamma_x^\dagger \Gamma_x K_{\text{BCS}}^{\text{no}} & 0 & 0 \\ 0 & 0 & \Gamma_y^\dagger \Gamma_y K_{\text{BCS}}^{\text{no}} & 0 \\ \Gamma_z^\dagger \Gamma_0 K_{\text{BCS}}^s & 0 & 0 & \Gamma_z^\dagger \Gamma_z K_{\text{BCS}} \end{pmatrix}. \quad (33)$$

There are off-diagonal components, which leads to the mixing of spin-singlet and spin-triplet superconductivities[25, 42]. However, numerical estimations show that these mixing is tiny for $J \lesssim t$. Hence, it is reasonable to assume that there is no mixing for $J \lesssim t$.

B. Superconducting pairing functions

The on-site attractive interaction U in Eq.(1) induces the spin singlet s -wave superconducting gap without the $\mathbf{k}$ dependence

$$\Delta_s(\mathbf{k}) = \Delta_s, \quad (34)$$

while the nearest-neighbor attractive interaction V in Eq.(1) may induce the spin-singlet extended s -wave superconducting gap

$$\Delta_{\text{ex-}s}(\mathbf{k}) = \Delta_{\text{ex-}s} (\cos ak_x + \cos ak_y), \quad (35)$$

or the spin-singlet d -wave superconducting gap

$$\Delta_d(\mathbf{k}) = \Delta_d (\cos ak_x - \cos ak_y), \quad (36)$$

or the spin-triplet chiral p -wave superconducting gap

$$\Delta_p(\mathbf{k}) = \Delta_p (\sin ak_x + i \sin ak_y). \quad (37)$$

The spin-triplet chiral p -wave superconductor is a topological superconductor, which hosts Majorana chiral-edge states.

We also study the superconductivity on the triangular lattice. The on-site attractive interaction U induces the spin singlet s -wave superconducting gap

$$\Delta_s(\mathbf{k}) = \Delta_s, \quad (38)$$

while the nearest-neighbor attractive interaction V may induce the spin-singlet extended s -wave superconducting gap

$$\Delta_{\text{ex-}s}(\mathbf{k}) = \Delta_{\text{ex-}s} \sum_j \cos(\mathbf{n}_j \cdot \mathbf{a}\mathbf{k}) \quad (39)$$

with

$$\mathbf{n}_j = \left(\cos \frac{2\pi j}{3}, \sin \frac{2\pi j}{3} \right), \quad (40)$$

or the spin-triplet chiral p -wave superconducting gap

$$\Delta_p(\mathbf{k}) = \Delta_p \sum_j \omega^j \sin(\mathbf{n}_j \cdot \mathbf{a}\mathbf{k}) \quad (41)$$

with $\omega \equiv e^{2\pi i/3}$ and Eq.(40), or the spin-singlet d -wave superconducting gap

$$\begin{aligned} \Delta_d(\mathbf{k}) = & \Delta_d \sum_j \cos ak_x - \cos \frac{ak_x}{2} \cos \frac{\sqrt{3}ak_y}{2} \\ & + i \sin \frac{ak_x}{2} \sin \frac{\sqrt{3}ak_y}{2}, \end{aligned} \quad (42)$$

or the spin-singlet f -wave superconducting gap

$$\Delta_f(\mathbf{k}) = \Delta_f \sum_j \sin(\mathbf{n}_j \cdot \mathbf{a}\mathbf{k}) \quad (43)$$

with Eq.(40). Which superconductivity is realized should be determined by numerically solving the linearized gap equation.

C. Destabilization of superconductivity

We derive the critical temperature Eq.(9) in the main text. The linearized gap equation reads

$$1 = V \sum_{\mathbf{k}} \frac{\sum_{s=\pm 1} \tanh \frac{\xi(\mathbf{k}) + sJf(\mathbf{k})}{2T}}{2\xi(\mathbf{k})}. \quad (44)$$

We expand it with $Jf(\mathbf{k})/2T_{\text{cr}}$,

$$\begin{aligned}
1 &= VN(0) \int \frac{d\Omega_{\mathbf{k}}}{S_{\text{FS}}} \int_0^{\omega_D} \frac{d\xi}{\xi} \\
&\quad \left[2 \tanh \frac{\xi}{2T_{\text{cr}}} - \left(\frac{Jf(\mathbf{k})}{2T_{\text{cr}}} \right)^2 2 \tanh \frac{\xi}{2T_{\text{cr}}} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}} \right] \\
&\simeq VN(0) \int_0^{\omega_D} \frac{d\xi}{\xi} \\
&\quad \left[2 \tanh \frac{\xi}{2T_{\text{cr}}} - \frac{J^2 \langle f(\mathbf{k})^2 \rangle_{\text{FS}}}{2T_{\text{cr}}^2} \tanh \frac{\xi}{2T_{\text{cr}}} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}} \right], \tag{45}
\end{aligned}$$

where we have used

$$\tanh(x \pm \delta) = \tanh x \pm \delta \text{sech}^2 x - \frac{\delta^2}{2} 2 \tanh x \text{sech}^2 x \tag{46}$$

and

$$\sum_{\pm} \tanh(x \pm \delta) = 2 \tanh x - 2\delta^2 \tanh x \text{sech}^2 x. \tag{47}$$

We define

$$F(T) \equiv UN(0) \int_0^{\omega_D} \frac{d\xi}{\xi} \tanh \frac{\xi}{2T}, \tag{48}$$

which satisfies the condition

$$F(T_{\text{cr}}^{(0)}) = 1. \tag{49}$$

We expand it as

$$F(T_{\text{cr}}) - 1 \simeq (T_{\text{cr}} - T_{\text{cr}}^{(0)}) F'(T_{\text{cr}}^{(0)}) = F'(T_{\text{cr}}^{(0)}) \delta T \tag{50}$$

with

$$\begin{aligned}
&F'(T_{\text{cr}}^{(0)}) \delta T \\
&= VN(0) \langle f(\mathbf{k})^2 \rangle_{\text{FS}} \int_0^{\omega_D} \frac{d\xi}{\xi} \frac{J^2}{2T_{\text{cr}}^2} \tanh \frac{\xi}{2T_{\text{cr}}} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}} \\
&\simeq VN(0) \langle f(\mathbf{k})^2 \rangle_{\text{FS}} \int_0^{\omega_D} \frac{d\xi}{\xi} \frac{J^2}{2T_{\text{cr}}^2} \tanh \frac{\xi}{2T_{\text{cr}}^{(0)}} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}^{(0)}}, \tag{51}
\end{aligned}$$

where

$$\langle f(\mathbf{k})^2 \rangle_{\text{FS}} \equiv \int_{\text{FS}} \frac{d\mathbf{k}}{S_{\text{FS}}} f(\mathbf{k})^2 \tag{52}$$

is the integration over the Fermi surface. Then, the δT is obtained as

$$\delta T = \frac{VN(0) \langle f(\mathbf{k})^2 \rangle_{\text{FS}} \int_0^{\omega_D} \frac{d\xi}{\xi} \frac{J^2}{2T_{\text{cr}}^2} \tanh \frac{\xi}{2T_{\text{cr}}^{(0)}} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}^{(0)}}}{F'(T_{\text{cr}}^{(0)})}, \tag{53}$$

where

$$\begin{aligned}
F'(T) &= VN(0) \int_0^{\omega_D} \frac{d\xi}{\xi} \left[\frac{\partial}{\partial T} \tanh \frac{\xi}{2T} \right] \\
&= -VN(0) \int_0^{\omega_D} d\xi \text{sech}^2 \frac{\xi}{2T_{\text{cr}}^{(0)}} d\xi. \tag{54}
\end{aligned}$$

δT is further calculated as

$$\delta T \equiv T_{\text{cr}} - T_{\text{cr}}^{(0)} = -C \langle f(\mathbf{k})^2 \rangle_{\text{FS}} \left(\frac{J}{T_{\text{cr}}^{(0)}} \right)^2. \tag{55}$$

C is calculated as

$$C \equiv \frac{\int_0^{\omega_D} \frac{\tanh \frac{\xi}{2T_{\text{cr}}^{(0)}} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}^{(0)}}}{\xi} d\xi}{-4 \int_0^{\omega_D} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}^{(0)}} d\xi} \tag{56}$$

$$\simeq \frac{-28\zeta'(-2)}{-\frac{1}{T_{\text{cr}}^{(0)}}} \simeq 28\zeta'(-2) T_{\text{cr}}^{(0)}, \tag{57}$$

where we have used

$$-4 \int_0^{\omega_D} \text{sech}^2 \frac{\xi}{2T_{\text{cr}}^{(0)}} d\xi = -\frac{\tanh \frac{\omega_D}{2T_{\text{cr}}^{(0)}}}{T_{\text{cr}}^{(0)}} \simeq -\frac{1}{T_{\text{cr}}^{(0)}} \tag{58}$$

for $\omega_D \gg 2T_{\text{cr}}^{(0)}$, and

$$\int_0^{\infty} \frac{\tanh \frac{\xi}{2} \text{sech}^2 \frac{\xi}{2}}{\xi} d\xi = -28\zeta'(-2), \tag{59}$$

where

$$\zeta'(x) \equiv \frac{d\zeta(x)}{dx} \tag{60}$$

is the derivative of the zeta function. Finally, we obtain the result

$$\delta T \simeq 28\zeta'(-2) \frac{J^2 \langle f(\mathbf{k})^2 \rangle_{\text{FS}}}{T_{\text{cr}}^{(0)}}, \tag{61}$$

which is Eq.(9) in the main text.

D. Tight-binding models

The tight-binding Hamiltonian is given by

$$H = H_{\text{Kine}} + H_X. \tag{62}$$

The kinetic term reads

$$H_{\text{Kine,Sq}} = 2t(2 - \cos ak_x - \cos ak_y) \tag{63}$$

on the square lattice, while it reads

$$H_{\text{Kine,Tri}} = 2t \left(3 - \cos ak_x - \sum_{\pm} \cos a \frac{k_x \pm \sqrt{3}k_y}{2} \right) \tag{64}$$

on the triangular lattice.

Tight-binding models for p -wave, d -wave, and g -wave altermagnets are identical to those in the previous literature[9, 21–23, 49]. The X -wave term f_X is given by

$$f_p = \sin ak_x \quad (65)$$

for the p -wave magnet[11, 12, 17, 20, 37, 45], by

$$f_d = 2(\cos ak_y - \cos ak_x) \quad (66)$$

for the d -wave magnet[2, 3, 26–28, 46–48], and by

$$H_g = 2 \sin ak_x \sin ak_y (\cos ak_y - \cos ak_z) \quad (67)$$

for the g -wave magnet[9, 21].

On the other hand, we use similar tight-binding models for f -wave and i -wave magnets in the present manuscript comparing with those in the previous literature[9, 21–23, 49].

1. f -wave magnets

We use the tight-binding model for the f -wave magnets,

$$f_f = 8 \left(-\sin ak_x + \sin a \frac{k_x + \sqrt{3}k_y}{2} + \sin a \frac{k_x - \sqrt{3}k_y}{2} \right). \quad (68)$$

It is constructed based on the nearest-neighbor sites. Note that it is similar to the tight-binding model used in the previous literature[9, 21–23, 49] but different, where the previous one is

$$f_f = 4 \left(\sin ak_x \sin a \frac{k_x + \sqrt{3}k_y}{2} \sin a \frac{-k_x + \sqrt{3}k_y}{2} \right), \quad (69)$$

which is constructed based on the third-nearest-neighbor sites.

2. i -wave magnets

We use the tight-binding model for the i -wave magnets,

$$\frac{2^{10}}{3\sqrt{3}} J\sigma_z \prod_{j=0}^2 \sin\left(\frac{a}{2} \mathbf{n}_j^A \cdot \mathbf{k}\right) \prod_{j=0}^2 \sin\left(\frac{\sqrt{3}a}{2} \mathbf{n}_j^B \cdot \mathbf{k}\right) \quad (70)$$

It is constructed based on the 4th-nearest-neighbor sites. It is the simplest tight-binding model with the i -wave symmetry because the 4th-nearest-neighbor sites are the minimum, which constitutes of the hopping of 12 identical sites. Note that it is similar to the tight-binding model used in the previous literature[9, 21–23, 49] but different, where the previous one is

$$16J\sigma_z \prod_{j=0}^2 \sin(a\mathbf{n}_j^A \cdot \mathbf{k}) \prod_{j=0}^2 \sin(\sqrt{3}a\mathbf{n}_j^B \cdot \mathbf{k}), \quad (71)$$

which is constructed based on the 12th-nearest-neighbor sites.

E. Topological numbers

1. Chern number

We show that the in-plane chiral ($p_x + ip_y$)-wave superconductor coupled with altermagnets is a topological superconductor. The BdG Hamiltonian reads

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \hat{\xi}(\mathbf{k}) & \hat{\Delta}(\mathbf{k}) \\ \hat{\Delta}^\dagger(\mathbf{k}) & -\hat{\xi}^T(-\mathbf{k}) \end{pmatrix}. \quad (72)$$

For the in-plane chiral p -wave superconductor in the presence of the altermagnetic order, it is explicitly given by

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \frac{\hbar^2 k^2}{2m} + Jf_X(\mathbf{k}) - \mu & \Delta_p(k_x + ik_y) \\ \Delta_p(k_x - ik_y) & -\frac{\hbar^2 k^2}{2m} - Jf_X(\mathbf{k}) + \mu \end{pmatrix} \quad (73)$$

in the continuum limit. We expand it in terms of the Pauli matrices as

$$H_{\text{BdG}}(\mathbf{k}) = \mathbf{h}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (74)$$

where

$$h_x(\mathbf{k}) = \Delta_p k_x, \quad h_y(\mathbf{k}) = \Delta_p k_y, \quad (75)$$

$$h_z(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} + Jf_X(\mathbf{k}) - \mu. \quad (76)$$

The Chern number is calculated as[50]

$$C = \frac{1}{4\pi} \int \mathbf{h}(\mathbf{k}) \cdot \left(\frac{\partial \mathbf{h}(\mathbf{k})}{\partial k_x} \times \frac{\partial \mathbf{h}(\mathbf{k})}{\partial k_y} \right) d^2k, \quad (77)$$

which gives $C = 1$. In the main text, we numerically evaluate the Chern number without taking the continuum limit.

2. Winding number

We show that the out-of-plane p_x -wave superconductor coupled with odd-parity magnets is a topological superconductor. The BdG Hamiltonian reads

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \hat{\xi}(\mathbf{k}) & \hat{\Delta}(\mathbf{k}) \\ \hat{\Delta}(\mathbf{k}) & -\hat{\xi}(\mathbf{k}) \end{pmatrix} = \hat{\xi}(\mathbf{k})\tau_z + \hat{\Delta}(\mathbf{k})\tau_x. \quad (78)$$

The BdG Hamiltonian has chiral symmetry defined by

$$\{\tau_y, H_{\text{BdG}}(\mathbf{k})\} = 0. \quad (79)$$

The chiral index is defined by

$$W \equiv \frac{1}{2\pi} \int_0^{2\pi} \tau_y H_{\text{BdG}}^{-1}(\mathbf{k}) \frac{\partial_{k_x} H_{\text{BdG}}(\mathbf{k})}{\partial k_x} dk_x. \quad (80)$$

By changing the basis, the BdG Hamiltonian is rewritten in the form of

$$\begin{aligned} H_{\text{BdG}}(\mathbf{k}) &= \hat{\xi}(\mathbf{k})\tau_x + \hat{\Delta}(\mathbf{k})\tau_y \\ &= \begin{pmatrix} 0 & \hat{\xi}(\mathbf{k}) - i\hat{\Delta}(\mathbf{k}) \\ \hat{\xi}(\mathbf{k}) + i\hat{\Delta}(\mathbf{k}) & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & q^*(\mathbf{k}) \\ q(\mathbf{k}) & 0 \end{pmatrix}. \end{aligned} \quad (81)$$

Then, the chiral index is identical to the winding number[44]

$$W(k_y) \equiv \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{q(\mathbf{k})} \frac{\partial q(\mathbf{k})}{\partial k_x} dk_x \quad (82)$$

with

$$q(\mathbf{k}) \equiv \hat{\xi}(\mathbf{k}) + i\hat{\Delta}(\mathbf{k}). \quad (83)$$

The winding number is nonzero if $q(\mathbf{k})$ encircles the origin.

3. *p*-wave magnets

Eq.(15) is rewritten as

$$q(\mathbf{k}) = c(\mathbf{k}) + M\mathbf{u}(\mathbf{k}) \quad (84)$$

with the center

$$c(\mathbf{k}) \equiv 2t \cos ak_y - \mu,$$

and

$$M \equiv \begin{pmatrix} 2t & J \\ 0 & \Delta_p \end{pmatrix}, \quad (85)$$

$$\mathbf{u}(\mathbf{k}) = \begin{pmatrix} \cos ak_x \\ \sin ak_x \end{pmatrix}. \quad (86)$$

The orbit of $q(\mathbf{k})$ is given by the ellipse satisfying $c(\mathbf{k}) + M\mathbf{u}$ with $|\mathbf{u}| = 1$. Then, the condition that the ellipse encircles the origin is

$$c(\mathbf{k}) + M\mathbf{u}(\mathbf{k}) = \mathbf{0} \quad (87)$$

for $|\mathbf{u}| < 1$. We solve $\mathbf{u}$ as

$$\begin{aligned} \mathbf{u}(\mathbf{k}) &= -M^{-1}c(\mathbf{k}) \\ &= \left\{ \frac{J\Delta_s}{2t\Delta_p} + \mu - \cos ak_y, -\frac{\Delta_s}{\Delta_p} \right\}. \end{aligned} \quad (88)$$

Hence, the condition is obtained as

$$\left(\frac{J\Delta_s}{2t\Delta_p} + \mu - \cos ak_y \right)^2 + \left(\frac{\Delta_s}{\Delta_p} \right)^2 < 1. \quad (89)$$

The winding number is 1 when the ellipse encircles the origin.

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